DS	**Session**	Graph	**Question Information**	Level 1 Challenge 88

Question description

You are given a tree consisting of n nodes. A *matching* is a set of edges where each node is an endpoint of at most one edge. What is the maximum number of edges in a matching?

Input Format

The first input line contains an integer n: the number of nodes. The nodes are numbered 1,2,…,n.

Then there are n−1 lines describing the edges. Each line contains two integers a and b: there is an edge between nodes a and b.

Output Format

Print one integer: the maximum number of pairs.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le a,b \le n$

Example

Input:

5

1 2

1 3

3 4

3 5

Output:

2

Explanation: One possible matching is (1,2) and (3,4)

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
1 2
1 3
3 4
3 5
```

EXPECTED OUTPUT

```
2
```

Test Case 2

INPUT (STDIN)

```
5
1 1
1 2
2 3
3 4
```

EXPECTED OUTPUT

```
6
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void dfs(int p,int i)
```

Test Case 2

KEYWORD

```
void link(int i,int j)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
9
```

Test Case 2

TOKEN COUNT

```
350
```

Test Case 3

NLOC

```
50
```

Code

tree_matching.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define MAXN 200005

typedef struct Edge {
    int to;
    struct Edge* next;
} Edge;

Edge* head[MAXN];
Edge pool[2 * MAXN];
int pool_ptr = 0;
bool matched[MAXN];
int max_matching = 0;

void link(int i, int j) {
    pool[pool_ptr].to = j;
    pool[pool_ptr].next = head[i];
    head[i] = &pool[pool_ptr++];

    pool[pool_ptr].to = i;
    pool[pool_ptr].next = head[j];
    head[j] = &pool[pool_ptr++];
}

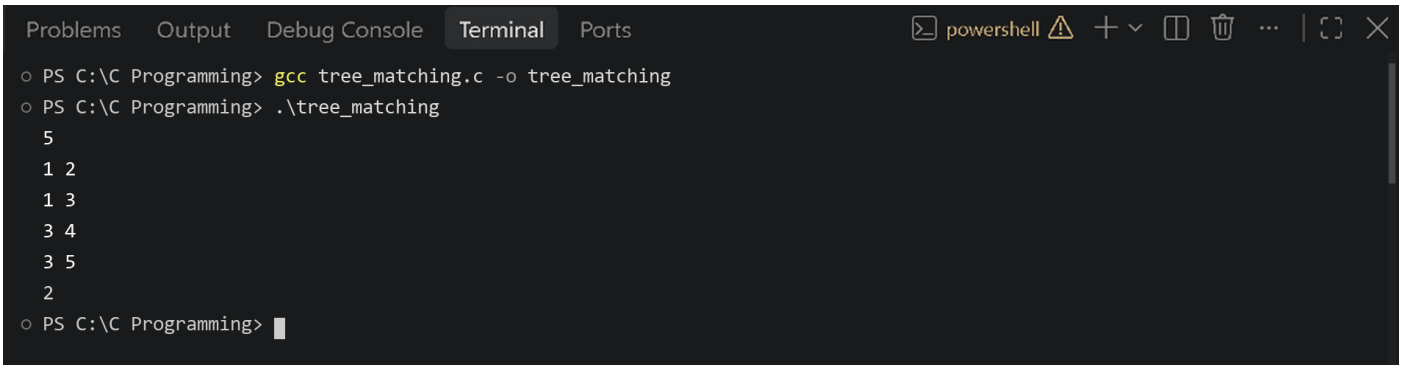
void dfs(int p, int i) {
    for (Edge* e = head[i]; e; e = e->next) {
        if (e->to != p) {
            dfs(i, e->to);
            if (!matched[i] && !matched[e->to]) {
                matched[i] = true;
                matched[e->to] = true;
                max_matching++;
            }
        }
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    for (int i = 0; i < n - 1; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        link(u, v);
    }

    dfs(0, 1);
    printf("%d\n", max_matching);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc tree_matching.c -o tree_matching
PS C:\C Programming> .\tree_matching
5
1 2
1 3
3 4
3 5
2
PS C:\C Programming>